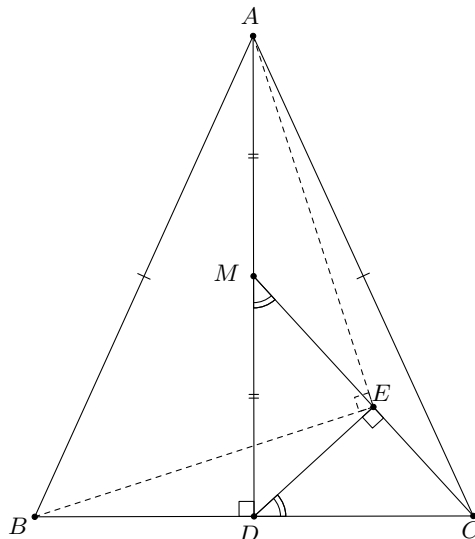


Problem 42. Let ABC be an acute-angled triangle with $AB = AC$. Suppose that D is the foot of the perpendicular from A to BC , M is the midpoint of AD , and E is foot of the perpendicular from D to CM . Prove that $\angle AEB = 90^\circ$.

Solution.



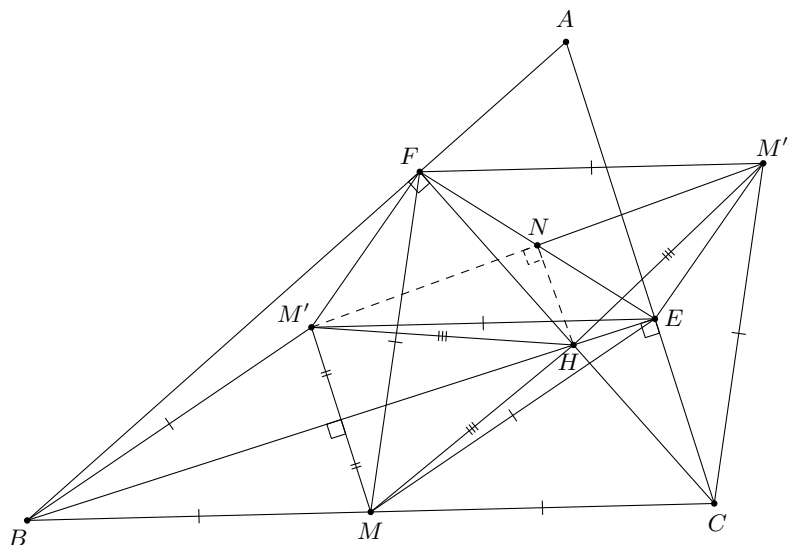
To prove that $\angle AEB = 90^\circ$, it is enough to show that $\triangle MEA \sim \triangle DEB$ because this implies that $\angle AEM = \angle BED$ and so $\angle AEB = \angle AEM + \angle MEB = \angle BED + \angle MEB = \angle MED = 90^\circ$. Now observe that $\angle AME = 180^\circ - \angle EMD = 180^\circ - \angle EDC = \angle BDE$, and, since $\triangle MDE \sim \triangle DCE$,

$$\frac{ME}{MA} = \frac{ME}{MD} = \frac{DE}{DC} = \frac{DE}{DB}.$$

□

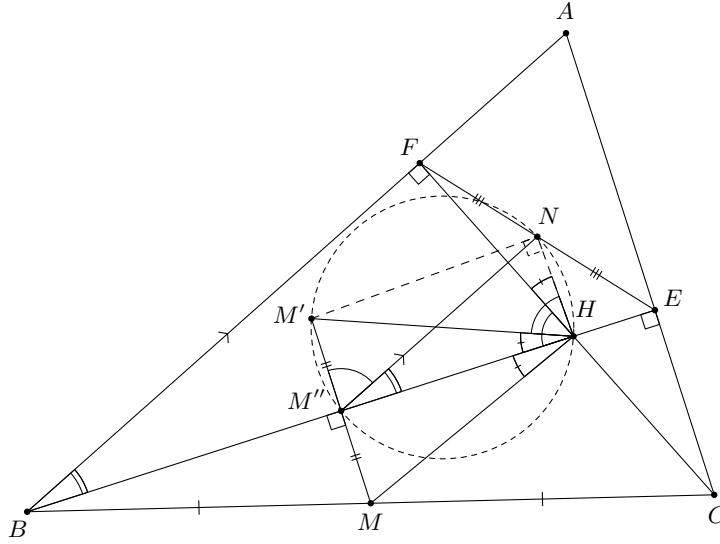
Problem 43. Let ABC be an acute-angled triangle with $AB > AC$ and orthocentre H . Suppose that E and F are the feet of the perpendiculars from B and C to CA and AB , respectively, M and N are the midpoints of BC and EF , respectively, and M' is the reflection of M about BE . Prove that $\angle M'NH = 90^\circ$.

Solution 1.



Since $MM' \parallel CA$, it follows that MM' bisects BE , and so the diagonals of $BMEM'$ bisect each other and are perpendicular to each other. This means that $BMEM'$ is a rhombus. Moreover, by symmetry, we have $HM = HM'$. Similarly, if M'' is the reflection of M about CF , then $CM''FM$ is a rhombus. Again, by symmetry, $HM = HM''$. Moreover, $M'E = ME = MC = M''C = FM''$ and $M'E \parallel MC \parallel FM''$. This implies that $FM'E M''$ is a parallelogram and N is the midpoint of MM'' . Since $HM' = HM = HM''$, it follows that $\angle M'NH = 90^\circ$. □

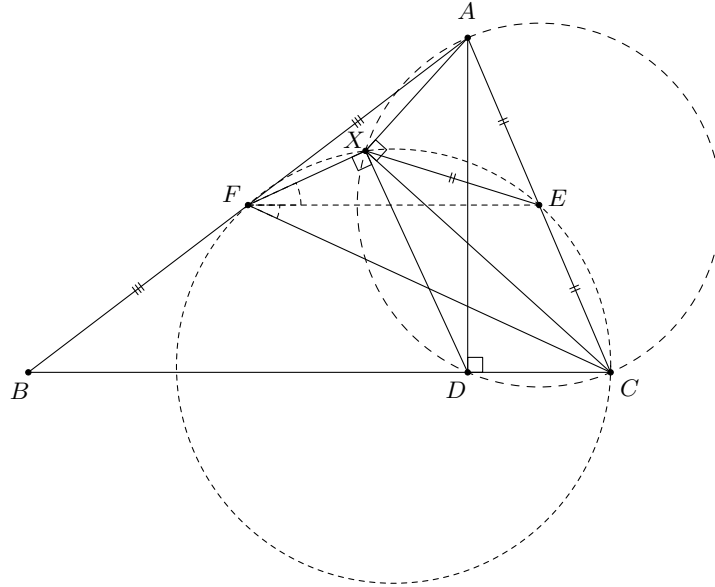
Solution 2.



Let $M'' = BE \cap MM'$. Since MM' bisects BE , it follows that $M''N$ is a midline in $\triangle BEF$, and so $\angle NM''H = \angle FBE = \angle FBH$. Hence $\angle M'M''N = 90^\circ - \angle NM''H = 90^\circ - \angle FBH = \angle FHB$. Since B, C, E, F are concyclic, it follows that $\triangle HFE \sim \triangle HBC$. So $\angle NHF = \angle MHB$ because these angles are formed by corresponding sides and corresponding medians. Hence $\angle M'HB = \angle MHB = \angle NHF$, and so $\angle NHM' = \angle FHB = \angle M'M''N$. Therefore M', M'', H, N are concyclic and so $\angle M'NH = 180^\circ - \angle M'M''H = 90^\circ$. $\square$

Problem 44. Let ABC be an acute-angled triangle with $AB > AC$. Suppose that D is the foot of the perpendicular from A to BC , E and F are the midpoints of CA and AB , respectively, and X is point inside $\triangle ABC$ such that $\angle CXA = \angle FXD = 90^\circ$. If X and B do not lie on the same side with respect to CF , prove that EF bisects $\angle CFX$.

Solution 1.



Since $\angle ADC = 90^\circ = \angle AXC$, it follows that A, X, D, C are concyclic, and so $\angle CXD = \angle CAD = 90^\circ - \angle ACB$. Hence

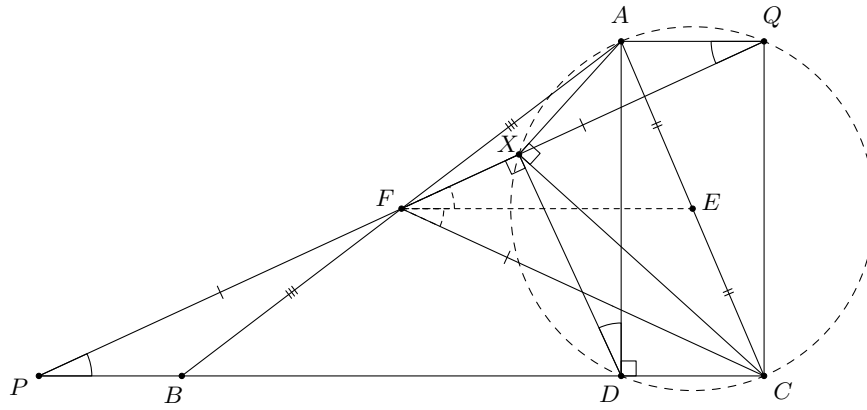
$$\angle CXF = \angle CXD + 90^\circ = 180^\circ - \angle ACB = \angle CEF$$

(because $EF \parallel BC$), and so F, C, E, X are concyclic. Therefore

$$\angle XFE = \angle XCE = \angle EXC = \angle EFC$$

(because $\triangle EXC$ is isosceles, since E is the centre of the circumcircle of $AXDC$) and so EF bisects $\angle CFX$. $\square$

Solution 2.



Let $P = XF \cap BC$ (which lies to the left of B in the configuration above). Let Q be the point to the right of A so that the quadrilateral $ADCQ$ is a rectangle. Since $\angle DXP = \angle ADP = 90^\circ$, it follows that $\angle ADX = \angle XPD$. Since $AXDCQ$ is cyclic, we have $\angle ADX = \angle AQX$ which means that $\angle AQX = \angle XPD$. Since $AQ \parallel BP$, it follows that Q, X and P are collinear. Since $AF = FB$ and $AQ \parallel BP$, we have $QF = FP$, and so, because $\angle QCB = 90^\circ$, it follows that $FC = QF = FP$. Hence $\angle XFC = 2\angle FPC$. Therefore EF bisects $\angle CFX$ because $FE \parallel BC$. $\square$